Report

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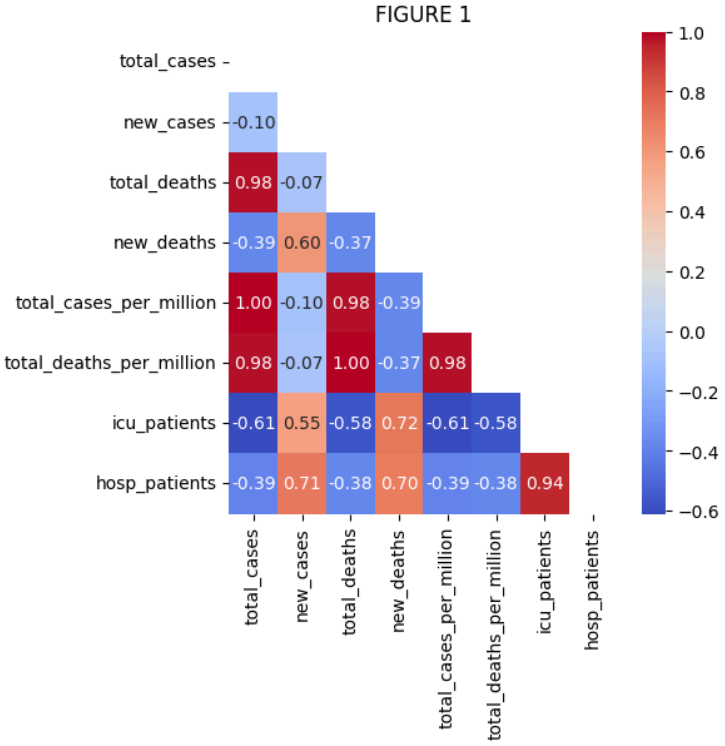
Covid provided many different datasets with information on several factors and general statistics of covid 19. This information can be used to predict future healthcare needs for pandemics and can reveal trends in the data that also help future preparations. This project seeks to answer the question of what the hospitalization and ICU needs were like during covid 19 and the trends of the pandemic. we. We utilize several of the case and death metrics to determine this along with several supervised and unsupervised techniques.I used correlation, outlier detection, regression, unsupervised and supervised learning techniques. Using these techniques we found which models and predictors were best for our data to predict the trend in hospitalization needs and icu requirements.

The first thing I did in figure 1 in my code was make a heatmap using the original variables only because the derived variables were made using the originals that will just create obvious trends and clutter in our map. Our heat map was rounded to two decimal places and showed us the correlations between variables and there are some interesting patterns we can see.

FIGURE 1

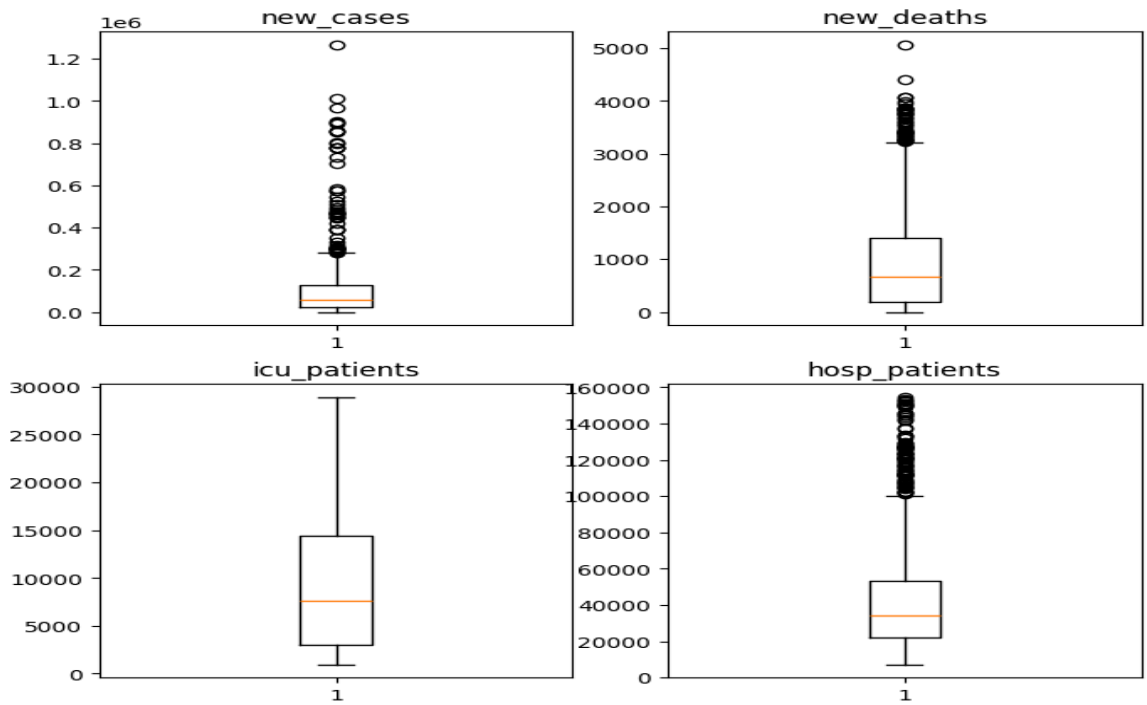


The first thing we need to think about is that some of the statistics only rise over time and they can not go down. Some examples of those variables are total cases and total deaths. This may cause some fluctuation in our correlations but not much but most of our original variables fluctuate. The first patterns we see are the obvious ones: total cases,total deaths, total cases per million, total deaths per million all have a strong

positive correlation. We also see that there is a strong positive correlation with hospital patients and icu patients but with all of the previous listed variables icu and hospital patients are negatively correlated my hypothesis is that as time went on the amount of people going to the hospitals and icu decreased either due to deaths or vaccines.

We can see from the figure below that the values from new_cases, new_deaths and hosp_patients have a lot of outliers. This is most likely due to the values that were recorded when data collection first started and when the outbreak first began and when the omnicorn variant emerged, hospitalizations surged along with deaths and cases. We also notice that there appear to be no outliers in the icu box plot and my hypothesis is that because the icus were at capacity for most of the pandemic no outliers can be found. If my hypothesis is correct then we can see that the United States clearly was not ready to handle a pandemic or national tragedy of this scale.

FIGURE 2



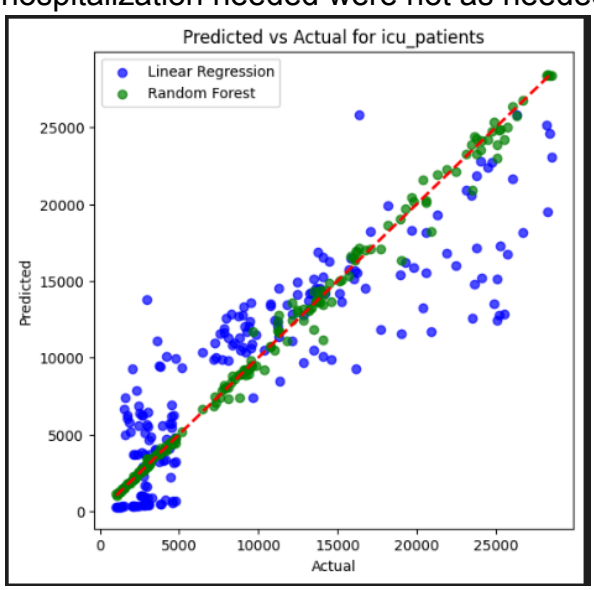
I ran linear regression and random forest and then I viewed several metrics from the regression models. I used icu_paitents and hosp_paitents as the two target metrics for our values. As for the predictors I used total_cases, new_cases, total_deaths and new_deaths and I used two regression models: linear regression and random forest. We know that covid did not follow a linear pattern so we can already assume that it will not perform great compared to random forest that adapts to the data. First for icu patients we got a r^2 score of .75 for linear and .99 for random forest. We know that we want our r^2 value to be as close to +1 as possible and that is the percent of variance in our data explained by the regression model. The random forest regression model performed very well with a mean absolute error of 3.14 and a root mean square error of

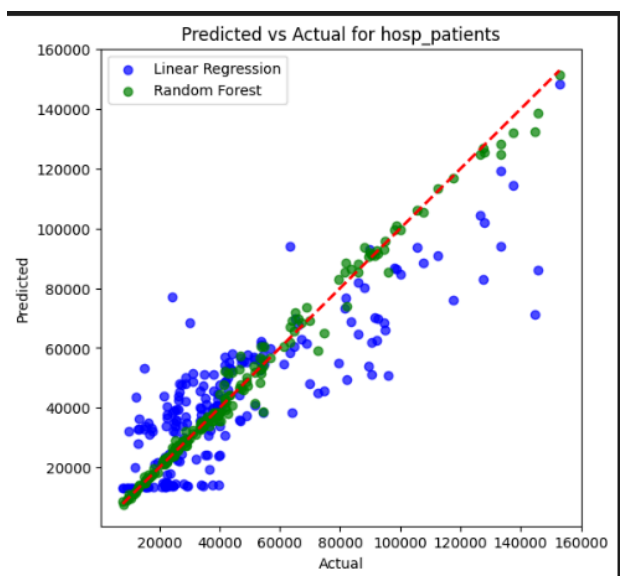
585 compared to the linear regressions 2954 and 3946 respectively. Mean absolute error takes the prediction minus the mean and the absolute value of that gives us a value for the error, once we have all those values we sum them and average to get the mean absolute error. The root mean square error takes the prediction minus the mean and squares it and takes the average but then takes the square root of the whole thing, the root mean square error is the more accurate value because it penalizes larger errors more. As for the hospital patients we had the same pattern: the random forest performed way better than the linear regression model.

```
_______TARGET_______: icu_patients
LINEAR
R^2: 0.75
MAE: 2954.71
RMSE: 3946.33

RANDOM FOREST
R^2: 0.99
MAE: 314.93
RMSE: 585.01
```

```
_______TARGET_______: hosp_patients
LINEAR
R^2: 0.71
MAE: 12528.95
RMSE: 16865.54

RANDOM FOREST
R^2: 0.99
MAE: 1885.70
RMSE: 3274.57
```

Our plots for icu and hospital show several extreme points. These are the points furthest away from the red dashed line we can see that the predictions on the lower end that were underestimated can hypothesize that those are from the early pandemic before people knew about covid or first learned. We also have extreme points at the end that were over estimated these are likely some time after the vaccine came out and icu and hospitalization needed were not as needed as much.

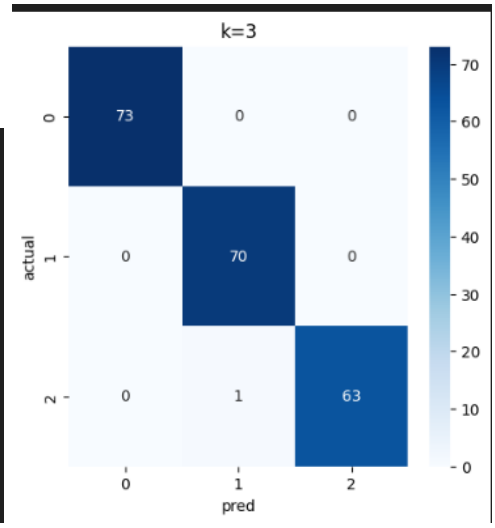




Next we are going to run the k nearest neighbors algorithm to predict the classes of our test data. We are going to use several variables from our data set to predict this. K nearest neighbors works by looking at the class value of some predetermined value k nearest neighbors and assigns based on what has the most votes. The variables we used predicted our test points extremely well meaning our chosen variables can predict what we should keep our eye on in future pandemics. So I ran k nearest neighbors

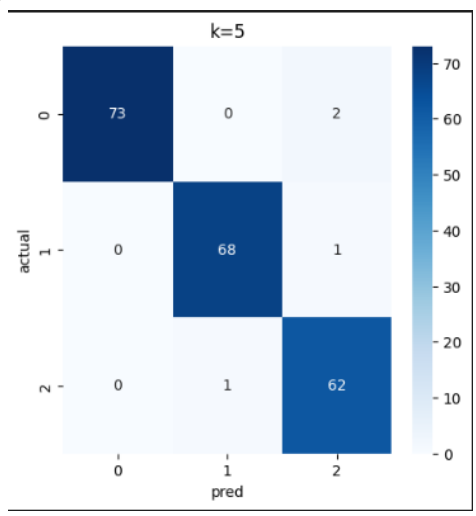
using the k values 1,3,5 and 10 I then did a classification report using the built in sk learn function. The value of k that performed the best was k=3 with an accuracy of almost 100%.

```
_____k=3_________
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        73
           1       0.99      1.00      0.99        70
           2       1.00      0.98      0.99        64

    accuracy                           1.00       207
   macro avg       1.00      0.99      1.00       207
weighted avg       1.00      1.00      1.00       207
```



As for our icu requirement our best value of k was k equals 5 with an accuracy of .98 this is slightly less accurate than the hospital predictions but still a great model when you consider the scale of the numbers in our data.

```
_____k=5_________
              precision    recall  f1-score   support

           0       1.00      0.97      0.99        75
           1       0.99      0.99      0.99        69
           2       0.95      0.98      0.97        63

    accuracy                           0.98       207
   macro avg       0.98      0.98      0.98       207
weighted avg       0.98      0.98      0.98       207
```



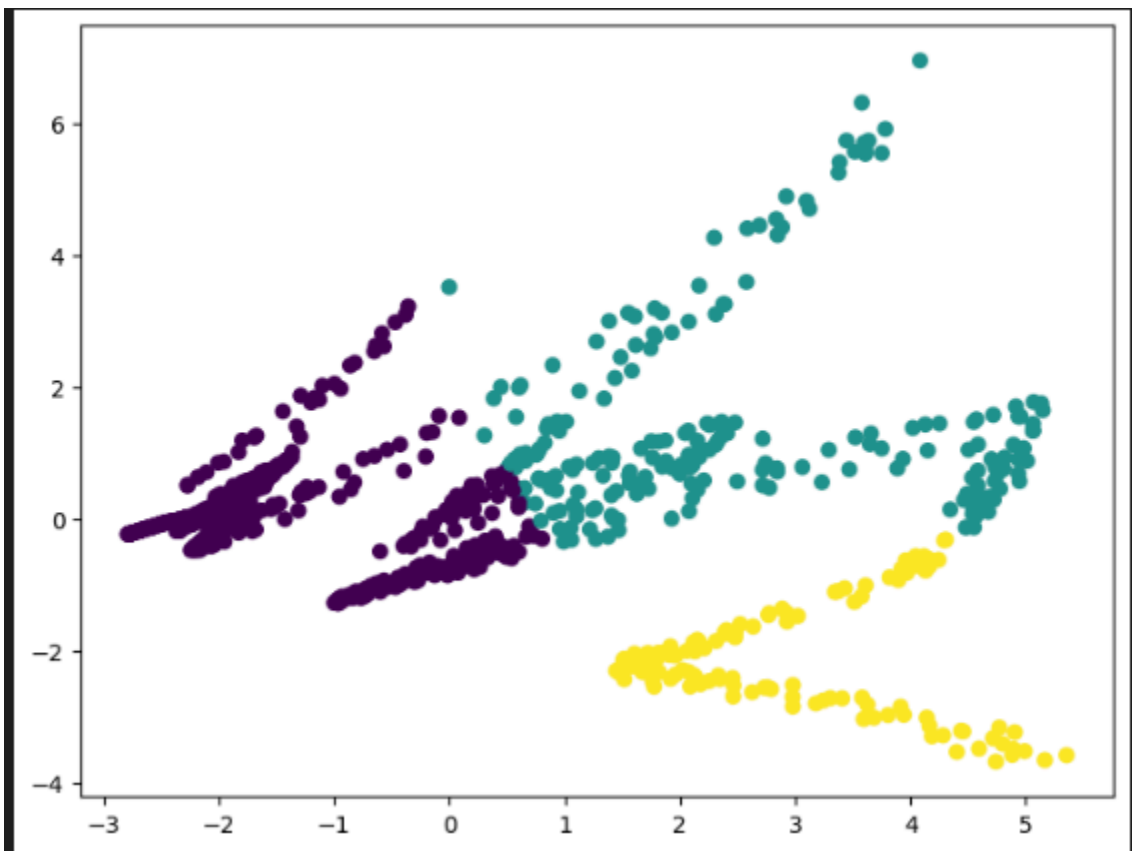
The ability of these variables to predict our data classes this well means these variables are something to be tracked in the event of a future pandemic and can help predict needs for the week, month or year. The variables I used that helped predict our data were things like hospital patients, hospitalization rate, and new_cases. This also proved that hospitalizations and ICU admissions were/are very predictable when we view the hospital and case metrics.

Next I will look at clustering using our data. I use almost all of the original variables and a couple of the derived ones. We classify the data using Kmeans and DBSCAN. Since clustering algorithms are distance based and we are using multiple dimension data with different scales we first scale our data then use the built in sklearn algorithm for DBSCAN and Kmeans. The way k means works is it randomly assigns

class values to our points and finds the average of all the dimensions this gives us the new centroid we then iterate and reassign and find new centroids till nothing changes in an iteration. As for DBSCAN we do not specify the number of clusters it determines based on the parameters given epsilon and min sample which are the distance a point needs to be to a core point to be considered a part of the cluster and the minimum number of neighbors needed to be considered a core point. In order to visualize this we need to perform a dimensionality reduction i used principal component analysis which lets us reduce our 8 predictors into a 2d map and our first two principal components returned are meaning 85% of the variance in our data is covered so the 2d model is almost completely correct with almost all of the data bering represented.

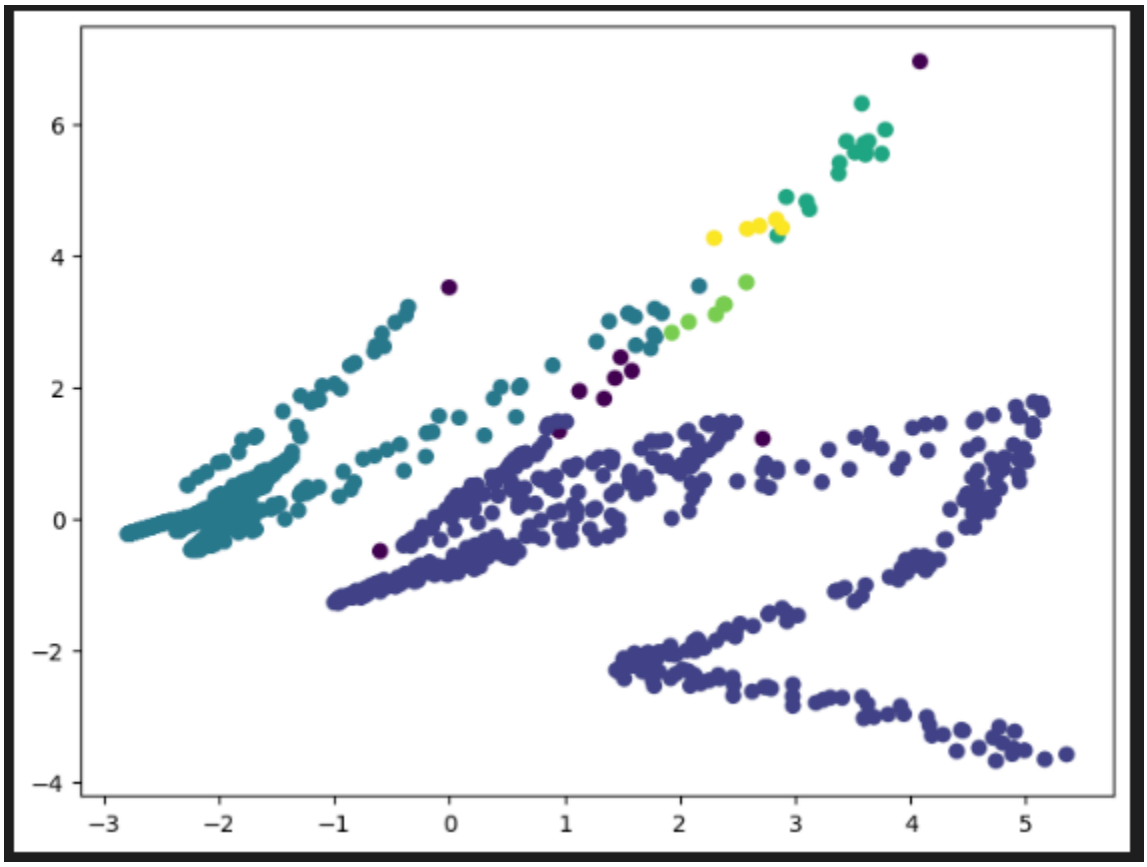
```
PC1 variance %: 61.846504680128
PC2 variance %: 23.68293807820461
```

cluster	0	1	2
hospitalization_need			
0	0.048295	0.565341	0.386364
1	1.000000	0.000000	0.000000
2	0.912023	0.087977	0.000000



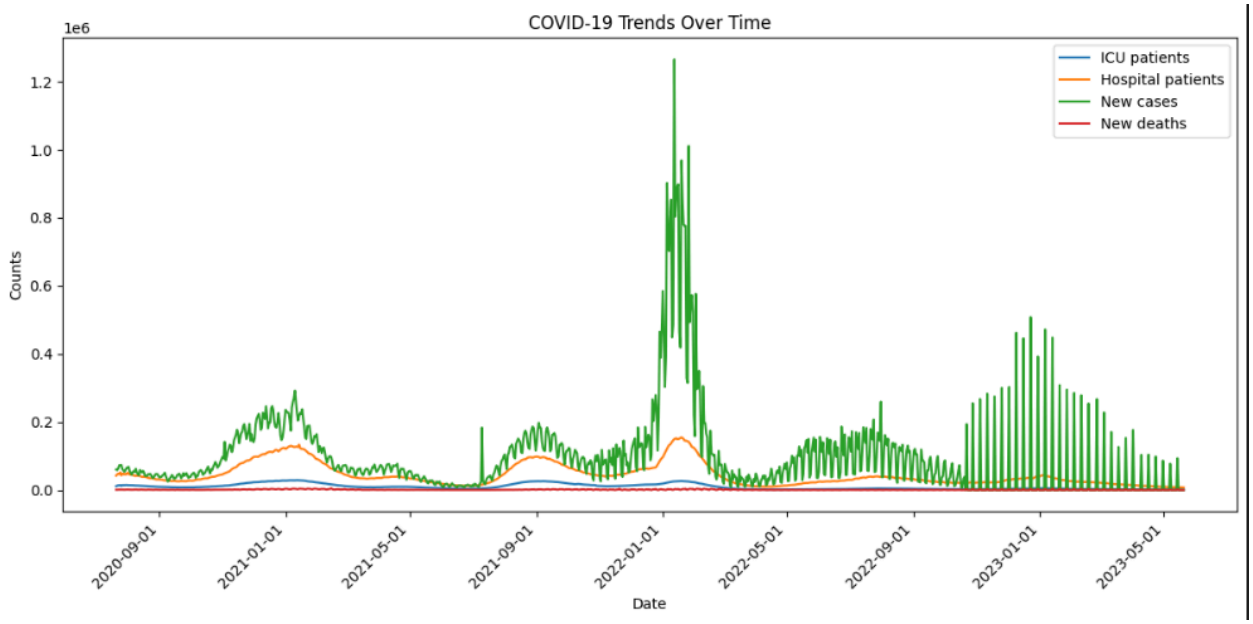
This is our plot for Kmeans after using pca to reduce the dimensions and based on the crosstab above we can see that all of the points in the medium class were assigned to the 0th cluster and we see that low was the hardest to classify being the most spread out. DBSCAN produced different results giving us the following information

```
cluster_db                 -1         0         1         2         3
hospitalization_need
0                    0.019886  0.880682  0.025568  0.042614  0.017045
1                    0.000000  0.000000  1.000000  0.000000  0.000000
2                    0.008798  0.621701  0.369501  0.000000  0.000000

cluster_db                  4
hospitalization_need
0                    0.014205
1                    0.000000
2                    0.000000
```



We can see that some of our classes are fitted into one cluster solely and we see again that the low class is the most unpredictable.

This proves that there is some meaningful statistic to look at when clustering our data. We see that most of the medium and high classes can be clustered into one single cluster for each of them and that means there are some patterns and predictability for our data. Cases for low are more variable when it comes to hospitalizations and harder to predict. Clustering is not the best choice to predict our data but this is one of the first things in a real world scenario that you would do most likely using DBSCAN or Kmeans and you would not have the labels provided for you.



Finally we have a simple plot I made that shows us key variables being plotted across all our dates. This visualizes the phases of covid perfectly. We can clearly see the moment that covid first started getting bad and then you can see when most people started getting the vaccine and then you can see the big spike that was when the new variation of the disease developed. The new variation and that big spike in our graph is the reason for a lot of the inaccuracies in our data. Those are the outlier points and part of the reason linear regression didn't work as well. This plot reinforces what we have learned using our models and proves our prediction that the covid data is not linear.

In conclusion we see covid-19 hospitalizations and icu requirements were highly predictable when using the appropriate machine -learning methods. We see that based on our data being nonlinear, linear regression did not work very well but random forest regression was able to capture this very well giving us a very high accuracy. Then through machine learning we saw that many variables like new_cases and new_deaths played a big role in predicting the trends in covid-19 hospitalizations and ICU requirements, also demonstrating the importance of recording these statistics. Clustering revealed that there is some pattern to the medium and high classes in hospitals and ICUs requirements. Finally we looked at a basic graph of the key variables in our data and the spikes and valleys confirm what we already knew about the real life events of covid-19 but also reinforce how our machine learning techniques can be used to predict and anticipate the health care needs before they are known.